MARKET POWER AND STRATEGY: UNDERSTANDING THE THEORY OF THE FIRM

PRESENTATION

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Key Topics:

- Perfect Competition
- Monopoly & Price Power
- Monopolistic Competition & Differentiation
- Oligopoly: Cartels &Price Leadership

Introduction to Market Structures

- The Spectrum of Competition: Markets in a modern mixed economy exist along a spectrum ranging from highly competitive markets to completely monopolized ones.

- Key Determining Factors: The type of market structure is determined by several important factors:

  - Number of Producers: This can range from a single seller (monopoly) to a large number of sellers (perfect competition).

  - Product Differentiation: Products may be identical (homogeneous) or differentiated with unique features and branding.

  - Control Over Price: The extent of market power a firm has in influencing or setting prices.

- Objective: To understand how different market structures address the three fundamental economic problems:
  What to produce, how to produce, and for whom to produce.

Perfect Competition – Meaning and Features

- Meaning:

Perfect competition is a market structure where many small firms sell identical products, and no single firm can influence the market price.

- Features:

○ Many Buyers and Sellers: No individual participant controls the market.

○ Homogeneous Product: All firms sell identical goods.

○ Perfect Information: Buyers and sellers have full knowledge of prices and quality.

○ Free Entry and Exit: Firms can easily enter or leave the market.

○ Price Takers: Firms accept the market price and face a horizontal demand curve.

Perfect Competition – Market Equilibrium Condition

- Short-Run Profit Maximization: A firm maximizes profit by producing where Marginal Cost (MC) = Price (P).

- Equilibrium Condition: P = MC

- Economic Logic:
  - If P > MC, produce more to increase profit.
  - If P < MC, produce less to avoid losses.

- Supply Curve: The rising portion of the MC curve above the shutdown point represents the firm's supply curve.

Perfect Competition – Long-Run Equilibrium

- Entry and Exit: In the long run, all costs are variable. Firms enter the industry if there are economic profits and exit if there are chronic loss

- The Zero-Profit Condition: The forces of competition push the price down to the point where it equals the minimum average cost.

- Long-Run Equilibrium Condition:
  - P = MC = minimum Long-Run Average Cost (AC).
- Economic Impact: In the long run, competitive firms earn a "normal" return on investment but zero "economic" profit

Perfect Competition – Efficiency and Criticism

- Allocative Efficiency (Pareto Efficiency): Achieved when no one can be made better off without making someone else worse off.

- Efficiency Condition: At equilibrium, MU = P = MC, meaning marginal benefit equals marginal cost.

- Criticisms:
  - Market Failures: Externalities and imperfect information reduce efficiency.
  - Income Inequality: Efficient outcomes may still create unfair wealth distribution.
  - Limited Innovation: Small firms may lack resources for major R&D.

Perfect Competition – The Demand Curve

- Horizontal Demand: A competitive firm is a price-taker, so it faces a perfectly elastic (horizontal) demand curve.

- Market vs Firm:
  - Industry demand (DD): Downward-sloping.
  - Firm demand (dd): Horizontal at market price.

- Marginal Revenue: For a perfect competitor, P = MR = AR.

Graph Location: Chapter 8, Page 150, Figure 8-1

Perfect Competition – The Shutdown Condition

- Problem: What if price (P) falls below average cost?

Graph Location: Chapter 8, Page 153, Figure 8-3

- Short-Run Strategy: Continue operating if revenue covers Variable Costs (VC), since Fixed Costs (FC) must be paid even at zero output.

- Shutdown Rule:
  - Stay Open: If P ≥ AVC
  - Shut Down: If P < AVC

- Shutdown Point: The lowest price where TR = VC.

- Condition: Shutdown occurs when P < AVC (or TR < VC).

Monopoly – Definition and Characteristics

- Definition: A market structure where a single firm is the only seller of a product with no close substitutes.

- Characteristics:
  - Single Seller: One firm supplies the whole market.
  - Barriers to Entry: Patents, legal rules, high costs, or resource control prevent new firms.
  - Price Maker: Faces a downward-sloping demand curve and decides price and output.
  - No Close Substitutes: The product is unique, giving strong market power.

Monopoly – Market Equilibrium Condition

- Profit Maximization: A monopolist maximizes profit where Marginal Revenue (MR) = Marginal Cost (MC).

- Price Determination:
  - First find output where MR = MC.
  - Then use the demand curve to determine the highest price consumers will pay (P*).

- Equilibrium Condition: MR = MC, and typically P > MC.

- Economic Profit: If P > AC, the monopolist earns long-run economic profit.

Graph Location: Chapter 9, Page 181, Figure 9-4

Monopoly – Marginal Revenue (MR)

- Meaning: Marginal Revenue (MR) is the additional revenue earned from selling one more unit of output.

- MR Curve: For a monopolist, MR < P because selling an extra unit requires lowering the price on all units sold.

- Elasticity Relationship:
  - MR > 0: When demand is elastic.
  - MR = 0: When demand is unit elastic (Total Revenue is maximum).
  - MR < 0: When demand is inelastic.

Graph Location: Chapter 9, Page 179, Figure 9-3

Monopoly – Formal Equations

- Total Revenue (TR): $TR = P * q$
- Marginal Revenue (MR): $MR = d(TR)/dq$
- Profit (π): $\pi = TR - TC$

- Maximization Condition:
  - $d\pi/dq = d(TR)/dq - d(TC)/dq = 0 \Rightarrow MR = MC$

- The Elasticity Equation: $MR = P(1 - 1/|e|)$, where e is the price elasticity of demand. This shows that a monopolist will never produce in the inelastic range of the demand curve

Monopoly – Efficiency and Criticism

- Social Cost: Monopoly creates inefficiency because P > MC, leading to misallocation of resources.

- Deadweight Loss: Output is restricted below the socially optimal level, causing a welfare loss.

- Static vs Dynamic Effects:
  - Static: Higher prices and lower output reduce consumer welfare.
  - Dynamic: Profits may encourage innovation (Schumpeterian view), but this is debated.

- X-Inefficiency: Lack of competition may reduce cost-minimizing incentives.

Graph Location: Chapter 9, Page 187, Figure 9-6

Monopolistic Competition – Definition and Characteristics

- Definition: A market structure between perfect competition and monopoly, where many firms sell differentiated products.

- Key Characteristics:
  - Product Differentiation: Goods are similar but not identical (e.g., brands, quality, design).
  - Many Sellers: Numerous small firms compete in the market.
  - Free Entry and Exit: Firms can enter or leave in the long run.
  - Some Market Power: Each firm can influence its price due to product uniqueness.

Monopolistic Competition – Short-Run Equilibrium

- Behavior: In the short run, the firm acts like a mini-monopolist with a downward-sloping demand curve.

- Equilibrium: Output is determined where MR = MC.

- Profit or Loss:
  - P > AC: Economic profit.
  - P < AC: Losses.

- Demand Curve: Relatively elastic due to many close substitutes.

Graph Location: Chapter 10, Page 199, Figure 10-2 (a)

Monopolistic Competition – Long-Run Outcome

- Adjustment: If firms earn profits, new firms enter, shifting demand left and making it more elastic.

- Long-Run Position: Entry continues until demand is tangent to the AC curve.

- Equilibrium Conditions:
  - P = AC (Normal profit)
  - MR = MC (Profit maximization)

- Result: P > MC and firms operate with excess capacity.

Graph Location: Chapter 10, Page 199, Figure 10-2 (b)

Monopolistic Competition – Efficiency and Debate

- Inefficiency: Since P > MC, some deadweight loss exists compared to perfect competition.

- Excess Capacity: Firms do not produce at minimum AC, leading to underutilized capacity.

- Variety Debate:
  - Criticism: Product differentiation may cause wasteful advertising and higher prices.
  - Defense: Consumers benefit from greater variety and choice.

Oligopoly – Meaning and Features

- Meaning: A market structure where a few large firms dominate the industry.

- Key Features:
  - Interdependence: Each firm’s decisions affect its rivals.
  - Few Sellers: Typically a small number of dominant firms.
  - Entry Barriers: High costs, economies of scale, or strong brand loyalty limit new entry.
  - Strategic Behavior: Firms consider rivals’ reactions (game theory).

Oligopoly – Behavioral Models

- No Single Theory: Oligopoly has no single equilibrium model because firms' decisions depend on rivals' reactions.

- Main Models:
  - Collusion (Cartels): Firms cooperate and act like a monopoly.
  - Price Leadership: One leading firm sets the price; others follow.
  - Kinked Demand Curve: Explains price rigidity.
  - Game Theory: Firms use strategic decision-making to anticipate rivals' moves.

Oligopoly – Collusion (Cartels)

Graph Location: Chapter 10, Page 201, Figure 10-3

- Definition: An agreement among firms to coordinate price and output to maximize joint profits.

- Outcome: A successful cartel behaves like a monopoly.

- Mechanism:
  - Industry output is restricted.
  - Price is set where industry MR = MC.
  - Profits are shared among members.

- Condition: Profit is maximized where ΣMR = ΣMC.

Oligopoly – Challenges to Cartels

- Incentive to Cheat: Members may secretly lower prices to gain extra market share.

- Instability: Cartels often fail due to:
  - Large Number of Firms: Harder to coordinate agreements.
  - Monitoring Problems: Difficult to detect cheating.
  - Legal Restrictions: Many countries ban cartels under antitrust laws (e.g., the Sherman Act).

- Example: OPEC is a well-known cartel attempting to control oil supply and prices.

Oligopoly – Price Leadership

- Definition: A form of "tacit collusion" where one dominant or highly efficient firm (the leader) sets the
price, and the smaller firms (the followers) match it.

- Characteristics:
  - Avoids the need for formal (and often illegal) agreements.
  - Reduces price uncertainty and avoids price wars.
  - The leader typically sets the price to maximize its own profit, taking into account the supply of
the followers.

- Equation: The leader’s demand = Market Demand – Supply of Follower Firms

Oligopoly – Kinked Demand Curve

- Theory: Proposed by Paul Sweezy to explain price rigidity in oligopoly.

Graph Location: Chapter 10, Page 203, Figure 10-4

- Assumptions:
  - If a firm raises price, rivals do not follow → demand is elastic above the kink.
  - If a firm cuts price, rivals follow → demand is inelastic below the kink.

- Result: A break (gap) in the MR curve allows MC to change within that range without affecting price or output.

Oligopoly – Game Theory Approach

- Framework: Studies strategic decisions of interdependent firms.

- Payoff Matrix: Shows each firm's profit based on its action and rivals' actions.

- Nash Equilibrium: No firm can gain by changing strategy while others keep theirs unchanged.

- Prisoner's Dilemma: Explains why firms may not cooperate, even when collusion is mutually beneficial.

Graph Location: Chapter 10, Page 203, Figure 10-4

Oligopoly – Efficiency and Criticism

- Deadweight Loss: Output is lower and price higher than the socially optimal level (P > MC).
- High Advertising Costs: Non-price competition may increase expenses without improving real value.
- Barriers to Innovation: Market dominance can limit entry of technologically superior firms.
- Defense: Large-scale production may reduce costs due to economies of scale.

Measuring Market Power – Concentration Ratios

- Four-Firm Concentration Ratio (CR4): Measures the percentage of total industry sales controlled by the four largest firms.

- Interpretation:
  - Perfect Competition: Ratio close to 0%.
  - Oligopoly: Usually above 40%.
  - Monopoly: 100%.

- Examples (U.S.):
  - Cigarettes – 95%
  - Automobiles – 85%
  - Computers – 45%

- Limitation: Often ignores international competition, which may reduce actual market power.

Graph Location: Chapter 10, Page 188, Figure 10-1

Measuring Market Power – Herfindahl-Hirschman Index (HHI)

- Definition: A concentration measure calculated by summing the squares of each firm's market share.

- Formula:

$HHI = (Share_1)^2 + (Share_2)^2 + \ldots + (Share_n)^2$

- Scale:
  - Monopoly: 10,000 (100^2)
  - Perfect Competition: Close to 0

- Use: Widely used by antitrust authorities to assess mergers and market concentration.

Strategic Interaction in Oligopoly

- Core Idea: Firms must consider rivals' reactions before making decisions.

- Interdependence: A price cut may trigger matching cuts, leading to a price war.

- Behavior Types:
  - Non-Cooperative: Firms act independently, often reducing prices.
  - Cooperative (Collusion): Firms coordinate to maximize joint profits like a monopoly.

Public Policies to Combat Market Power

- Purpose: To reduce inefficiencies of imperfect competition, such as high prices (P > MC) and restricted output.

- Regulatory Tools:
  - Economic Regulation: Controls prices and entry, especially for natural monopolies.
  - Antitrust Policy: Prevents anti-competitive practices and excessive market concentration.

- Key U.S. Laws:
  - Sherman Act (1890): Bans restraints of trade.
  - Clayton Act (1914): Prohibits practices that reduce competition (e.g., price discrimination).

Summary – Comparison of Market Structures

- Perfect Competition: Many firms, identical products, no market power (P = MC), efficient outcome.

- Monopolistic Competition: Many firms, differentiated products, limited market power, zero profit in the long run.

- Oligopoly: Few dominant firms, high entry barriers, strategic behavior, possible long-run profits.

- Monopoly: Single seller, no close substitutes, strong barriers, price maker (P > MC), deadweight loss.

Conclusion – The Modern Mixed Economy

- Balanced Approach: A healthy economy needs competitive markets for efficiency and innovation, along with government intervention to correct market failures.

- Key Lessons:
  - Markets work best when competition exists and externalities are absent.
  - Monopoly power reduces consumer welfare through higher prices and lower output.
  - Strategic interaction defines competition in oligopolistic markets.